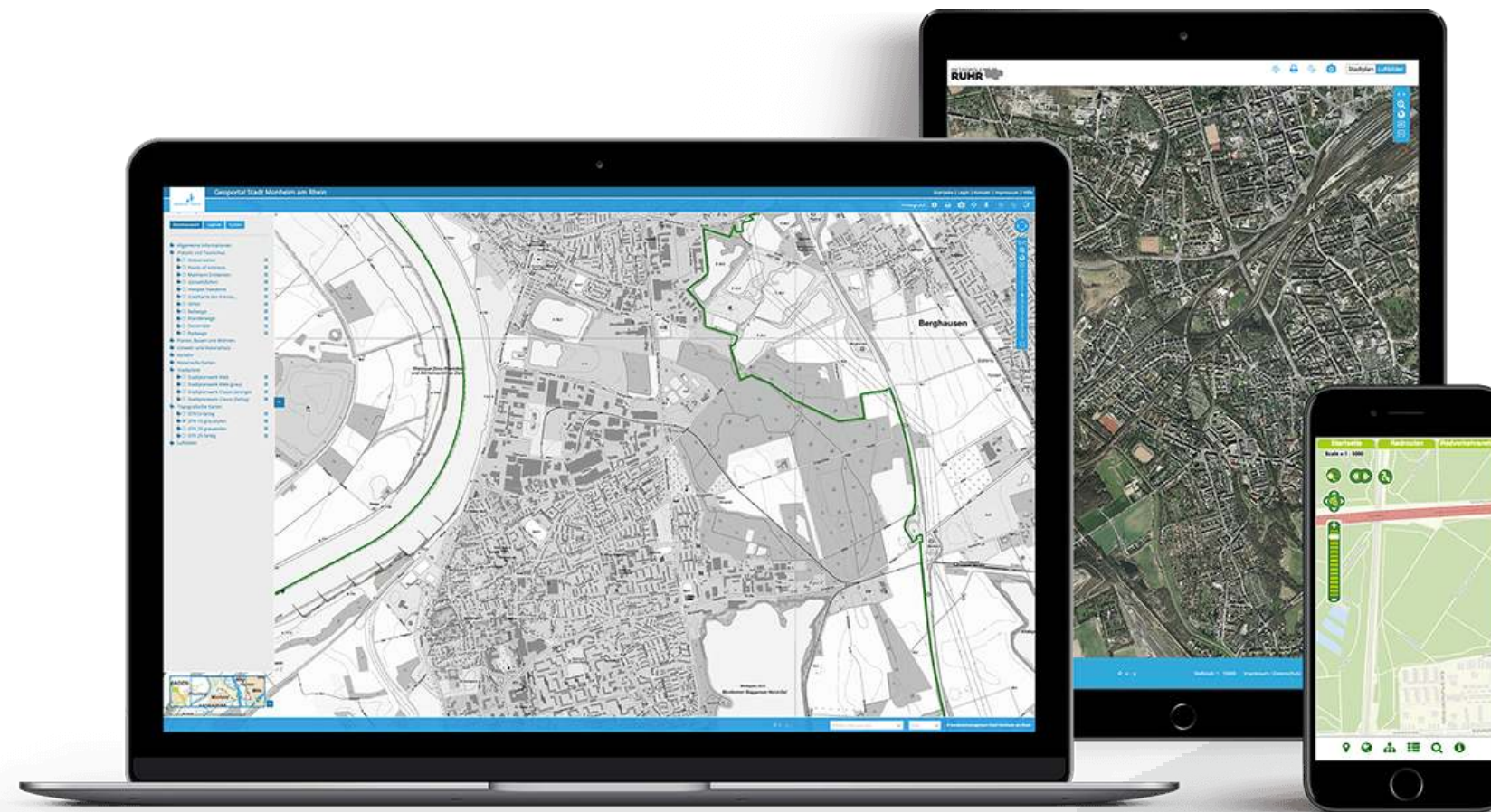


Create great Web Applications with Mapbender

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FOSS4G Europe 2026 Slides: <https://talks.osgeo.org/foss4g-europe-2026-workshops/talk/NZ7U8B/>

Astrid Emde

WhereGroup Bonn since 2002

focus on WebGIS & Webmapping consultancy &
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PostgreSQL/PostGIS, MapServer, GeoServer, QGIS,
QGIS Server, Mapbender, PostNAS-Suite

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> 50 employees

developers, consultants & geospatial experts

Bonn – Berlin - Freiburg - Hamburg

Open Source Software Solutions since > 15 years

Company behind Mapbender, MapComponents & Mops

FOSS Academy - WhereGroup Training Institute

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<https://wherogroup.com>

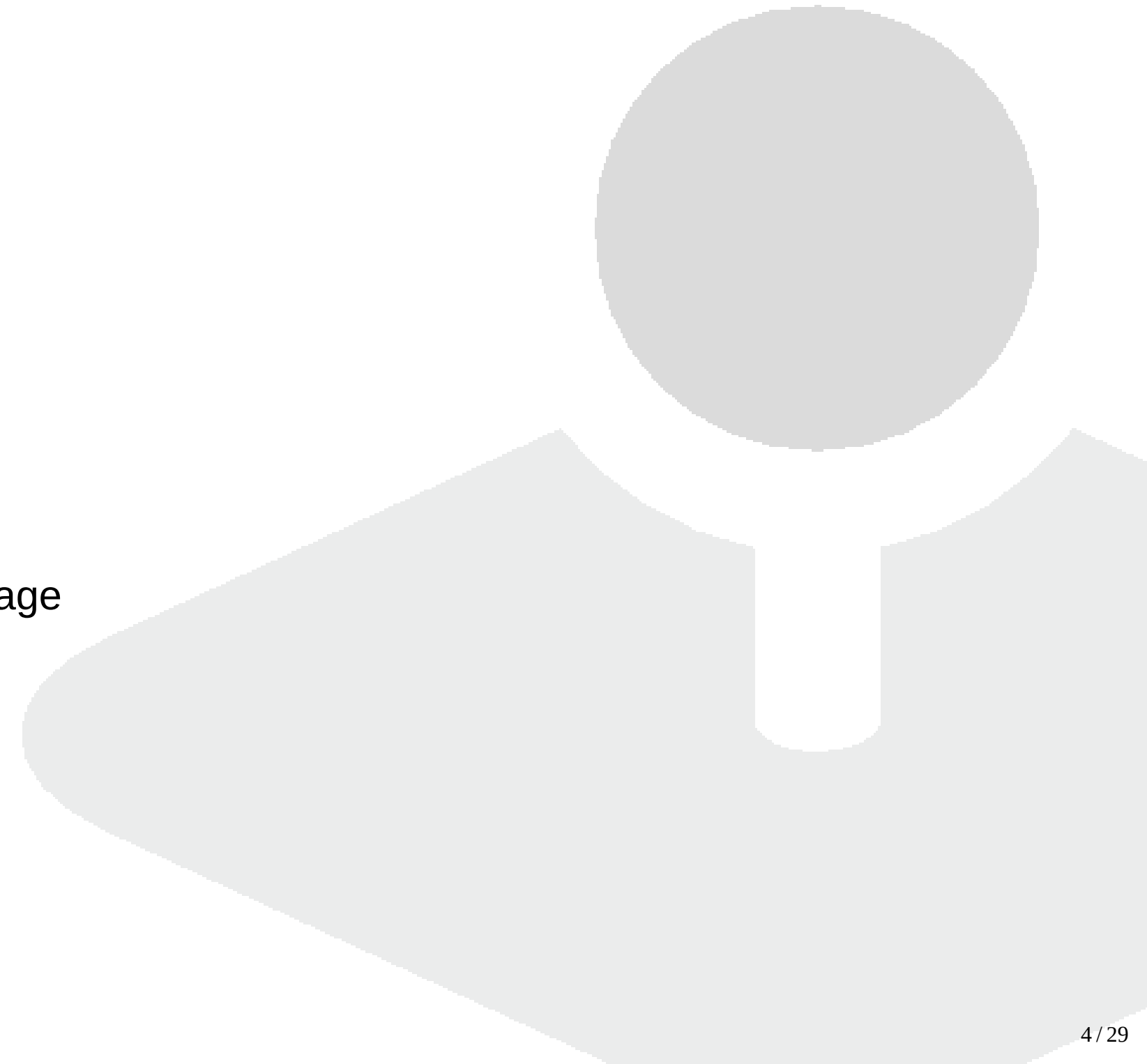
<https://foss-academy.com>



Mapbender

Easy to configure!

- WebGIS Client Suite with administration web interface
- Create a geoportal without writing a single line of code!
- Any number of applications with only one installation
- Create and maintain an OWS repository
- Distribute configured services among applications
- Grant users and groups access to applications and services
- Supports several languages and you can support your language
- <https://mapbender.org/en/about-mapbender/>



Mapbender

- OSGeo project since 2006
- MIT License
- Code on GitHub



Mapbender

The Architecture

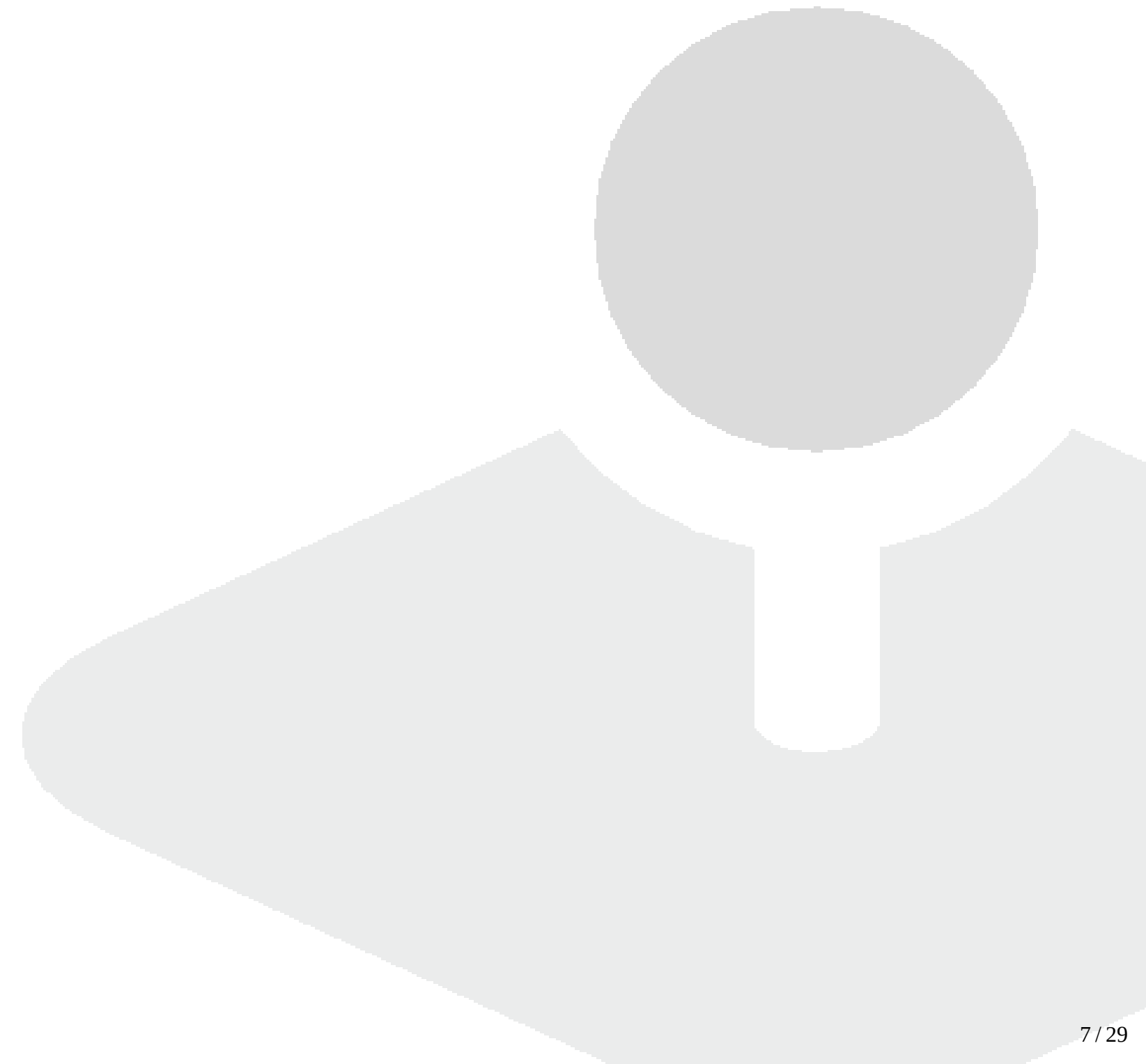
- PHP Framework Symfony
- HTML
- Bootstrap
- OpenLayers as map client
- YAML configuration



Mapbender

Requirements for your installation

- Running on your server
- Needs a WebServer like Apache HTTP Server or NGINX
- Needs PHP
- Database for administration i.e. SQLite or PostgreSQL



Mapbender Functionality

Offers many features that you can combine.

- Offers elements to visualize, create and edit data.
- Search & print functionality & much more
- <https://mapbender.org/en/features/>

Mapbender Gallery

Get inspired

- <https://mapbender.org/en/showcase/>

Mapbender

Applications

Offers template applications

- Template applications demonstrate the functionality
- Create a copy & build your own application
- <https://demo.mapbender.org/>

Mapbender 5.0.0-RC1

26. June 2026

- Final Version 5.0.0 expected in July 2026
- <https://github.com/mapbender/mapbender-starter/blob/master/CHANGELOG.md>
- <https://mapbender.org/aktuelles/details/mapbender-version-500-released/>
- <https://develop.mapbender.org>

Mapbender

Important Links

- Mapbender Website <https://mapbender.org>
- Mapbender Funktionalität <https://mapbender.org/funktionen/>
- Mapbender Dokumentation <https://doc.mapbender.org>
- Mapbender Project on Github <https://github.com/mapbender/>
- Mapbender-Starter on Github <https://github.com/mapbender/mapbender-starter/>
- Mapbender Workshop-Repository <https://github.com/mapbender/mapbender-workshop/>
- OSGeoLive <https://live.osgeo.org>
- Mapbender on OSGeoLive https://live.osgeo.org/de/quickstart/mapbender_quickstart.html

What we will learn

- Get to know Mapbender
- Add Services to Mapbender
- Create your own application
- Get to know Mapbender functionality
- Learn about Roles and permission handling

Open the Mapbender Landing page & login

- <https://schulung.foss.academy/mapbender/>
- Login via: root / foss4ge2026
- Explore Applications

Publish WMS in Mapbender

- Menu → Sources → + Add Source
- OGC WMS: Load GetCapabilities Url

Load Services in Mapbender

- FOSS4G WMS <https://wms.wherogroup.com/cgi-bin/foss4g.xml?SERVICE=WMS&VERSION=1.3.0&REQUEST=getCapabilities>
- OSM by WhereGroup <http://osm-demo.wherogroup.com/service?REQUEST=GetCapabilities&Service=WMS&Version=1.3.0>
- TopPlus Open https://sgx.geodatenzentrum.de/wms_topplus_open?request=GetCapabilities&service=wms
- WMS Mundialis <https://ows.mundialis.de/services/service?Service=WMS&Version=1.3.0&REQUEST=getCapabilities>
- WMS IGME5000 Geological Map <https://services.bgr.de/wms/geologie/igme5000/?REQUEST=GetCapabilities&VERSION=1.3.0&SERVICE=WMS>
- Mapbender at the Geological Institute of Romania <https://mapbender.igr.ro/application/mapviewerGIR>
- WMS Romania Meteorology <https://geoportal.meteoromania.ro/GIS/wms?Service=WMS&Version=1.3.0&Request=getCapabilities>
- WMS Romania Geology <https://inspire.igr.ro/geoserver/geolro200k/wms??Service=WMS&Version=1.3.0&Request=getCapabilities>
- OGC API - Features pygeoapi project <https://demo.pygeoapi.io/master>

Create a new Application in Mapbender

- Your application should start with the extent of Timisoara
- Copy Application mapbender_user and rename it to FOSS4G
- Switch to Layout → map-Element and set SRS EPSG:4326
- Modify MAX EXTENT to the whole world
lower left -180 -90
upper right 180 90
- Modify Start Extent Timisoara
lower left 21.173132 45.709645
upper right 21.274413 45.78900
- Modify the scales to:
100000000,50000000,10000000,5000000,1000000,500000,100000,50000,25000,10000,7500,5000,2500,1000,500
- Add other EPSG:31700, EPSG:3844
- Check for EPSG-Codes for Români <https://epsg.io/?q=Romania>

Add Services to your application

- Your application should show some OGC WMS
- Go to Tab Layerset
- Add WMS via +
- Create a new Layerset via "big" + and add Services to the new layerset
- Go to Layout to the **map** Element and activate your new Layerset and move it to the right position
- Go to Layout to element Layertree and open the folder for your new Layerset
- Have a look at your application. The new Layerset should show up

Modify your application Layout

Change and add elements

- Learn about Mapbender Elements <https://doc.mapbender.org/en/elements.html>
- Disable elements you do not want to provide
- Change the visibility for desktop and mobile use
- Change the behaviour of the sidepane
- Change the size of the element **Overview**

Modify your application Layout

Change and add elements

- Add more functionality
- Add a new element **Link** and refer to the FOSS4G website <https://2026.europe.foss4g.org> - see <https://doc.mapbender.org/en/elements/misc/link.html>
- Add a new element **HTML** to the toolbar that shows the application name - see <https://doc.mapbender.org/en/elements/misc/html.html>
- Add a new element **HTML** to the sidepane and write an Impressum where you add the Link <https://2026.europe.foss4g.org> and refer to the image <https://2026.europe.foss4g.org/images/logo-negru.png>

Modify your application Layout

Change and add elements

- Change the behaviour of the element **Layertree** - see <https://doc.mapbender.org/en/elements/basic/layertree.html>
- Change the behaviour of the element **Navigation Tolbar** - see https://doc.mapbender.org/en/elements/basic/navigation_toolbar.html

Mapbender PDF Print

Change and add elements

- Change the element **Print** - see <https://doc.mapbender.org/en/elements/export/printclient.html>
- Add scales, change the file prefix to foss4g
- Add a new PDF template **a4portrait_foss4g** (the template was prepared for FOSS4G and is on the server)
- Try the new Batch Print element - see <https://doc.mapbender.org/en/elements/export/printclientbatch.html>
- Print generate frames along a line - get GeoJSON from <https://github.com/astroindex/presentations/tree/master/FOSS4G-Europe-2026/>

Mapbender Design

Overwrite images and css

- Modifications in parameters.yaml - logo, name, splashscreen, login image
- Overwrite favicon mapbender/application/public/
- See css Editor in your application
- Change the toolbar button color
- See https://github.com/mapbender/mapbender/blob/master/docs/elements/css_custom_properties.md

```
:root {  
  --primary: #ff0000;  
  --button-background: #00ff00;  
  --toolbar-background: #0000ff;  
  --toolbar-text-color: #ff00ff;  
  --background-color: #ffff00;  
  --toolbar-top-background: #00ffff;  
  --toolbar-border-color: #ff0000;  
}
```

User and Access Management

Create User and Groups

- Go to Security
- Users → + Add new user - add two new users
- Group → + Add new group - add a group
- Group → add your first user to your new group

User and Access Management

Provide access to your application

- Go to your application to Security
- Grant view access to your new group
- Go to the Layout section and grant access to the Routing element only to your second user

Search

Search on your data in PostgreSQL

- Copy the searchRouter configuration from https://schulung.foss.academy/mapbender/application/mapbender_digitize_demo_foss4g to your application
- Have a look at the configuration
- https://doc.mapbender.org/en/elements/search/search_router.html

Digitizer

Create/Modify data via Mapbender

- Copy the digitizer configuration from https://schulung.foss.academy/mapbender/application/mapbender_digitize_demo_foss4g to your application
- Have a look at the configuration
- Create some new features
- Print via digitizer using your a4portait_foss4g template
- https://doc.mapbender.org/en/elements/editing/digitizer/digitizer_configuration.html

What is coming next?

More things to discover

- Join the mailing list
- Try out more elements
- Translate Mapbender to your language
- Try the QGIS Pugin QGIS2Mapbender to publish WMS with QGIS Server directly from QGIS - see <https://github.com/WhereGroup/qgis2mapbender/blob/main/README.md>
- Join the Community Sprint at FOSS4G Europe on Saturday & get involved.
- Reach out to me for more information

Thank you

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mapbender-users@list.osgeo.org



FOSS4G Europe 2026 Programm: <https://talks.osgeo.org/foss4g-europe-2026-workshops/talk/NZ7U8B/>

Mapbender Workshop Slides: <https://talks.osgeo.org/foss4g-europe-2026-workshops/talk/NZ7U8B/>

